EML HW 7

① a) $B(\phi-\theta) = B_m \cos(\phi-\theta) = B_m \cos(\phi-\omega t)$

b) $dA = l r \, d\phi$

$\Rightarrow$ flux $= \int_0^{\pi} B \cdot dA = 2 B_m l r \sin\theta$

flux linkage $\lambda = N\phi = 2NB_m l r \sin\theta$

$e = \frac{d\lambda}{dt} = 2NB_m l r \omega \cos\omega t$

② $e_1 = 2NB_m l r \omega \cos\omega t$

$\phi_2 = \int_{\delta}^{\pi+\delta} B \cdot dA = 2 B_m r l \sin(\theta-\delta)$

$\Rightarrow e_2 = 2NB_m l r \omega \cos(\omega t - \delta)$

$e = e_1 + e_2 = 4NB_m l r \omega \cos\frac{\delta}{2} \cos(\omega t - \frac{\delta}{2})$

③ $V = 208V$ $P_{out} = 10hp$ $p = 4$ $f = 60Hz$ $S = 0.05$

a) $n_{sync} = \frac{120 f}{P} = 1800$ rpm

b) $n_{motor} = (1-S) n_{sync} = 1710$ rpm

c) $f_{rotor} = S f_{electrical} = 3Hz$

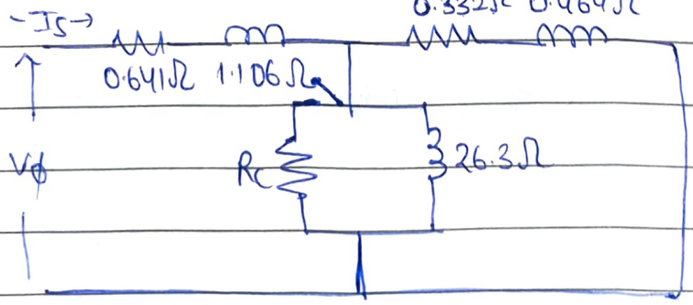
d) $P_{out} = 746.10 = 7460W$; $\omega_r = 1710 \times \frac{2\pi}{60} = 179.07$ rad/s

$T_{load} = \frac{P_{out}}{\omega_r} = 41.67$ Nm

④ $V = 460V$ $P = 25hp$ $p = 4$ $f = 60$ $S = 0.022$

0.332Ω 0.464Ω

$V_\phi = \frac{460}{\sqrt{3}} = 265.58$



a) $n = \frac{120 f}{P} = 1800$ rpm $\Rightarrow n_{motor} = (1-S)1800 = 1760.4$ rpm

b) $Z_2 = \frac{R_2'}{S} + j X_2' = 15.091 + 0.464j$

$Z_1 = R_1 + jX_1 = 0.641 + 1.106j$

$Z_{eff} = Z_1 + \frac{(jX_m) Z_2}{jX_m + Z_2} = 11.058 + 6.692j \Rightarrow |Z| = 14.06\Omega$, $\angle Z = 33.69°$

$I_s = \frac{265.58}{Z_{eff}} = 18.89 \angle -33.69$

c) P.F = cos(33.69) = 0.832 lagging

d) $R_{mech} = R_2'\left(\frac{1-s}{s}\right) = 14.759\Omega$

(5) 75KW ; 3.3KV ; 6 pole ; Y

a) No load: f = 50Hz ; V = 3.3KV ; I = 5A ; P = 2500W

Blocked rotor: f = 15Hz , V = 400V ; I = 27A ; P = 15KW

$3 \times 27^2 \times R_{eq} = 15 \times 10^3 \Rightarrow R_{eq} = 6.859\Omega$

$R_1 = 3.75\Omega \Rightarrow R_2' = 3.109\Omega$

$Z_{eq} = \frac{400/\sqrt{3}}{27} = 8.553 \Rightarrow X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = 5.110\Omega$

$(X = 2\pi f L)$

But $X_{eq} = 5.110$ at $f = 15Hz$, motor runs at 50Hz $\Rightarrow 5.110 \times \frac{50}{15} = 17.033\Omega$

$Z_{NL} = \frac{V}{I} = \frac{3300/\sqrt{3}}{5} = 381.05\Omega$

$R_{NL} = \frac{P}{3I^2} = \frac{2500}{3 \cdot 5^2} = 33.33\Omega \Rightarrow X_{NL} = 379.59\Omega$

b) $Z_2 = \frac{R_2'}{s} + jX_2'$ $\quad Z = Z_1 + \frac{(jX_m)(Z_2)}{jX_m + Z_2} = 75.07 + 31.44j$

$I_s = \frac{1905.26}{|Z|} = 23.41A$ $\quad PF = \cos 22.22 = 0.922$ lagging

$P_{in} = 3V_\phi I \cdot \cos\phi = 3(1905.26)(23.41)(0.922) = 123380W$

$P_{Cu} = 3(23.41)^2(3.75) = 6185W$

$\Rightarrow P_{airgap} = P_{in} - P_{Cu} = 117215W$

$P_{mech} = (1-s)P_{airgap} = 112526W$

$P_{out} = P_{mech} - P_{core} = 112526 - 2500 + 3(5)^2(3.75) = 110307W$

$\eta = \frac{110307}{123380} = 0.894W$

EML HW 8

① $n_s = \frac{120f}{P} = 3000$ rpm

a) $S = \frac{n_s - n_m}{n_s} = \frac{50}{3000} = 0.01667 = 1.67\%$

b) $\omega_m = \left(\frac{2\pi}{60}\right)(2950) = 308.92$ rad/s

$T = \frac{P_{out}}{\omega_m} = 48.56$ N/m

c) $T \propto$ slip $\Rightarrow S_{new} = 0.0333$

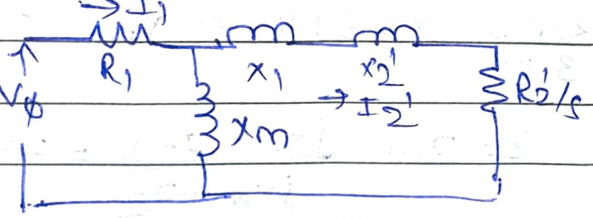
$\Rightarrow n_{new} = (1 - 0.0333)3000 = 2900$ rpm

d) $\omega_{new} = \frac{2900 \times 2\pi}{60} = 303.68$ rad/s

$\Rightarrow T = \frac{P}{\omega} \Rightarrow P = 2 \times 48.55 \times 303.68 = 29487$ W

② $V_\phi = 265.58$V ; $n_{sync} = \frac{120 \times 60}{4} = 1800$ rpm $= 188.5$ rad/s

a)



max power transfer theorm:

$\left|\frac{R_2'}{S}\right| = |R_1 + j(X_1 + X_2')|$

$\Rightarrow S_{max} = \frac{0.382}{\sqrt{0.641^2 + (1.106 + 0.464)^2}} = 0.1957$

$\Rightarrow N_r = (1 - S)N_S = 1447.6026$ rpm

$I_2' = \frac{V_\phi}{\sqrt{(R_1 + R_2/S)^2 + (X_1 + X_2')^2}} = 94.336$A

$\Rightarrow T_{max} = \frac{(I_2')^2(R_2'/S)}{\omega} = 240.1$ Nm

b) $S = 1 \Rightarrow I_2' = \frac{V_\phi}{\sqrt{(R_1 + R_2')^2 + (X_1 + X_2')^2}} = 143.7859$

$\Rightarrow T = \frac{(143.79)^2(0.332)}{(2\pi/180)(1800)} = 109.2422$

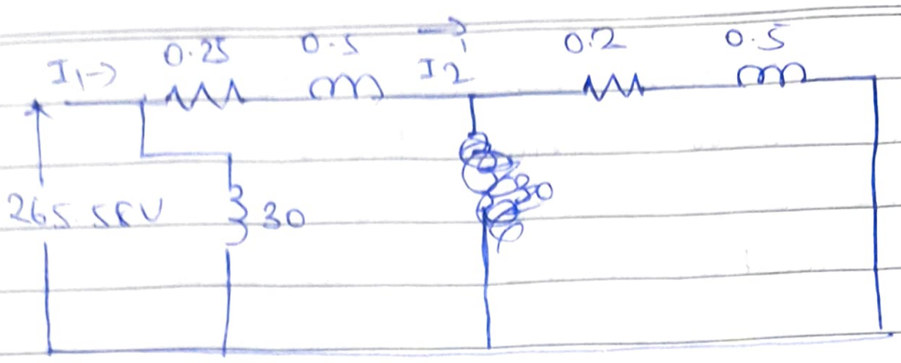
c) $R_2' = 0.664 \Rightarrow (S_2)_{max} = 0.3915 \Rightarrow N_r = (1 - S)1800 = 1095.2$

$I_2' = 130.09$A $\Rightarrow T = \frac{P_{80}}{2\pi \times 1800} \times 130.088^2 \times 0.664 = 178.839$ Nm

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(3)



$N_S = 1800$ rpm

a) $S = 1$

$$I_1 = \frac{265.58}{jX_m \,||\, [(R_1 + R_2') + j(X_1 + X_2')]} = 250 \angle -66.6^\circ \text{ A}$$

$$I_2' = \frac{265.58}{(R_1 + R_2') + j(X_1 + X_2')} = 243.19 \angle -65.77^\circ \text{ A}$$

$$T_{starting} = \frac{(243.19)^2 (0.2)}{\frac{2\pi}{180}(1800)} = 186.7 \text{ Nm}$$

b) $N_r = 1740$ rpm $\Rightarrow S = \frac{1800 - 1740}{1800} = 0.0333$

$$I_{1\,full} = \frac{V_\phi}{\left[(R_1 + R_2'/S) + j(X_1 + X_2')\right] \,||\, jX_m} = 44.230 \angle -20.49^\circ \text{ A}$$

$PF = \cos(-20.49^\circ) = 0.937$ lagging

(4) $N_S = \frac{120 \times 60}{6} = 1200$ rpm $\quad V_\phi = 265.58$ V

$R_1 = 0.2\,\Omega$, $R_2' = 0.2\,\Omega$ $\quad T \propto \frac{R_2}{S} \Rightarrow R_{2\,new}' = 0.2 \frac{(1200 - 1000)}{(1200 - 1140)}$

$= 0.6667$

$\Rightarrow R_{ext} = 0.4667\,\Omega$

(5) $T_{in} = 162 + 13.5 = 175.5$ NM

$f_r = 1.5$ HZ $\quad f_s = 50$ HZ $\Rightarrow S = \frac{f_r}{f_s} = 0.03$

$N_S = \frac{120 f}{P} = 1000$ rpm $\Rightarrow N_r = (1 - S) N_S = 970$ rpm

$P_o = 162 \times \left(\frac{2\pi}{60} \times 970\right) = 16455.662$ w

$P_{ag} = 175.5 \times \left(\frac{2\pi}{60} \times 1000\right) = 18378.319$ w

$P_{ag} : P_{rcl} = 1 : S \Rightarrow P_{rcl} = 551.349$ W

$P_{in} = P_{ag} + P_{\text{Stator}} = 750 + 18378.317 = 19128.317$ W

$\Rightarrow \eta = \frac{P_o}{P_i} \times 100 = 86.027\%$

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(6) $P_i = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 400 \times 50 \times 0.8 = 27712.812\,W$

$P_{SCL} = 1500W \qquad P_{core} = 1200W$

$P_{rCL} = 900W \qquad P_f = 1050W$

$P_{ag} = P_{in} - P_{SCL} - P_{core} = 25012.8129W$

(7)

$P_{BR} = \sqrt{3} \times 230 \times 140 \times 0.35 = 19520.2\,W \Rightarrow$

$P_{SCL} = P_{RCL} = 9760.1\,W$

$3 I_{BR}^2 (R_1 + R_2') = P_{BR}$

$\Rightarrow R_1 - R_2' = \frac{P_{BR}}{6 I_{BR}^2} = 0.16598\,\Omega$

$|Z_{BR}| = \left| \frac{V_{BR}/\sqrt{3}}{I_{BR}} \right| = \sqrt{(R_1 + R_2')^2 + (X_1 + X_2')^2}$

$\Rightarrow X_1 = X_2' = 0.4425\,\Omega$

$S_{max} = \frac{R_2'}{(R_1^2 + (X_1 + X_2')^2)^{1/2}} = 0.1836$

$T_{max} = \frac{180}{2\pi N_S} (I_2')^2 \frac{R_2'}{S_{max}}$

$T_{max} = \frac{|V_{\phi\,new}'|}{(R_1 + R_2'/S_{max})^2 + (X_1 + X_2')^2} \times \frac{R_2'}{S_{max}}$

$T_{max} = 300\,NM \Rightarrow V_{\phi\,new} = 149.690\,V$

$V_{L-L\,new} = \sqrt{3}\, V_\phi = 259.271\,V$

$\Rightarrow$ % red in voltage $= 41.074\%$

(8) $N_S = \frac{120 \times 50}{8} = 750\,RPM \qquad S_{max} = \frac{750 - 630}{750} = 0.16$

$\frac{R_2'}{S} = const \Rightarrow R_{ext} = R_{2\,new}' - R_{2\,old}' = 0.3675\,\Omega$

(9) $P_o = 50 \times 745.7 = 37.285\,KW \Rightarrow P_i = 41.427\,KW$

$P_{SCL} = P_{rCL} = P_{core} \qquad P_{no\,load} = 4$

$\Rightarrow P_{mech} = 4/3$

$P_{no\,load} = P_{core} + P_{mech} \Rightarrow P_{core} = \frac{2y}{3}$

$P_i = P_o + P_{SCL} + P_{rCL} + P_{core} + P_{mech} \Rightarrow P_i - P_o = \frac{2y}{3} \Rightarrow y = 1775.5$

$P_{ag} = P_i - P_{core} - P_{SCL} = P_i - \frac{4y}{3}$

$\frac{P_{ag}}{P_{rCL}} = \frac{1}{S} \Rightarrow S = 0.0303$

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(10) $\frac{R_2'}{S_{max}} = \sqrt{R_1^2 + (X_1 + X_2')^2}$ $R_1 \rightarrow$ negligible

$\Rightarrow$ $S_{max} = \frac{1}{3}$

$f_r = \frac{1}{3} \times 50 = 16.67\ Hz$

$f_s = f_r$ @ max to make max torque @ starting

$\frac{V_\phi}{f_s} = \frac{V_\phi\ new}{f_s\ new} \rightarrow V_{\phi\ new} = \frac{400 \times f_s/3}{f_s} = 133.33V$